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Shear-assisted extrusion assemblies and methods

Abstract

Shear-assisted extrusion assemblies are provided. The assemblies can include a billet containing assembly containing a billet comprising a billet outer material and a billet inner material in at least one cross-section; a tool operably engaged with the billet; an extrudate receiving channel configured to receive extrudate from the tool, wherein the extrudate comprises extruded outer material and extruded inner material in at least one cross-section, the extruded outer material being the same material as the billet outer material, and the extruded inner material being the same as the billet inner material. Methods for producing multi-material shear-assisted extrudate are also provided.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3432369	12/1968	Naastepad	N/A	N/A
3640657	12/1971	Rowe et al.	N/A	N/A
3661726	12/1971	Denes	N/A	N/A
3684593	12/1971	Benz et al.	N/A	N/A
3884062	12/1974	Green	N/A	N/A
3892603	12/1974	Reid	N/A	N/A
3933536	12/1975	Doser et al.	N/A	N/A
3977818	12/1975	Sprankle	N/A	N/A
3977918	12/1975	Paladino et al.	N/A	N/A
3989548	12/1975	Morris	N/A	N/A
4287749	12/1980	Bachrach et al.	N/A	N/A
4300378	12/1980	Thiruvarduchelvan	N/A	N/A
4431467	12/1983	Staley et al.	N/A	N/A
4585473	12/1985	Narasimhan et al.	N/A	N/A
4778542	12/1987	Clemens	N/A	N/A
4801340	12/1988	Inoue et al.	N/A	N/A
4808224	12/1988	Anderson et al.	N/A	N/A
4892596	12/1989	Chatterjee	N/A	N/A
4985085	12/1990	Chatterjee	N/A	N/A
5026438	12/1990	Young et al.	N/A	N/A
5089060	12/1991	Bradley et al.	N/A	N/A
5226989	12/1992	Sukonnik	N/A	N/A
5242508	12/1992	McCallum et al.	N/A	N/A
5262123	12/1992	Thomas et al.	N/A	N/A
5283130	12/1993	Bradley et al.	N/A	N/A
5437545	12/1994	Hirai	N/A	N/A
5461898	12/1994	Lessen	N/A	N/A
5470401	12/1994	McCallum et al.	N/A	N/A
5492264	12/1995	Wadleigh	N/A	N/A
5737959	12/1997	Korbel et al.	N/A	N/A
5739498	12/1997	Sunamoto et al.	N/A	N/A
5964117	12/1998	Holroyd et al.	N/A	N/A
5988484	12/1998	Osborn et al.	N/A	N/A

6022424	12/1999	Sellers et al.	N/A	N/A
6036467	12/1999	Jameson	N/A	N/A
6638462	12/2002	Davidson et al.	N/A	N/A
6843405	12/2004	Okamoto et al.	N/A	N/A
6940379	12/2004	Creighton	N/A	N/A
7096705	12/2005	Segal	N/A	N/A
7314670	12/2007	Bartsch et al.	N/A	N/A
7322508	12/2007	Chang et al.	N/A	N/A
7954692	12/2010	Fukuda	N/A	N/A
8016179	12/2010	Burford	N/A	N/A
8240540	12/2011	Tanaka et al.	N/A	N/A
8313692	12/2011	Somekawa et al.	N/A	N/A
8695868	12/2013	Messer et al.	N/A	N/A
10189063	12/2018	Lavender et al.	N/A	N/A
10369748	12/2018	Whalen et al.	N/A	N/A
10695811	12/2019	Joshi et al.	N/A	N/A
10987754	12/2020	Eller et al.	N/A	N/A
11045851	12/2020	Joshi et al.	N/A	N/A
2002/0029601	12/2001	Kwok	N/A	N/A
2004/0057782	12/2003	Okamoto et al.	N/A	N/A
2004/0238501	12/2003	Kawazoe et al.	N/A	N/A
2004/0265503	12/2003	Clayton et al.	N/A	N/A
2005/0081594	12/2004	Segal	N/A	N/A
2006/0005898	12/2005	Liu et al.	N/A	N/A
2006/0027628	12/2005	Sutherlin et al.	N/A	N/A
2008/0029581	12/2007	Kumagai et al.	N/A	N/A
2008/0048005	12/2007	Forrest et al.	N/A	N/A
2008/0202653	12/2007	Ignberg	N/A	N/A
2008/0251571	12/2007	Burford	N/A	N/A
2009/0072007	12/2008	Nagano	N/A	N/A
2009/0269605	12/2008	Warke	228/104	B21C 23/22
2009/0291322	12/2008	Watanabe et al.	N/A	N/A
2010/0059151	12/2009	Iwamura et al.	N/A	N/A
2010/0089976	12/2009	Szymanski et al.	N/A	N/A
2010/0132430	12/2009	Tsai et al.	N/A	N/A
2011/0104515	12/2010	Kou et al.	N/A	N/A
2011/0132970	12/2010	Nakagawa et al.	N/A	N/A
2011/0309131	12/2010	Hovanski et al.	N/A	N/A
2012/0006086	12/2011	Manchiraju et al.	N/A	N/A
2012/0052322	12/2011	Hatakeyama et al.	N/A	N/A
2012/0168045	12/2011	Hara et al.	N/A	N/A
2012/0223451	12/2011	Hulseman et al.	N/A	N/A
2012/0258332	12/2011	Hatakeyama et al.	N/A	N/A
2013/0075452	12/2012	Burford	N/A	N/A
2013/0266467	12/2012	Manchiraju et al.	N/A	N/A
2014/0000332	12/2013	Wilson et al.	N/A	N/A
2014/0002220	12/2013	Johnson et al.	N/A	N/A
2014/0076957	12/2013	Sayama et al.	N/A	N/A
2014/0102159	12/2013	Denison	N/A	N/A
2014/0102161	12/2013	Stewart	N/A	N/A

2014/0248508	12/2013	Ohhama et al.	N/A	N/A
2014/0260489	12/2013	Funk et al.	N/A	N/A
2014/0283574	12/2013	Lavender	72/262	B21C 23/002
2014/0291886	12/2013	Mark et al.	N/A	N/A
2014/0328710	12/2013	Cui et al.	N/A	N/A
2015/0075242	12/2014	Eller et al.	N/A	N/A
2015/0115019	12/2014	Pascal et al.	N/A	N/A
2015/0360317	12/2014	Kalvala et al.	N/A	N/A
2016/0008918	12/2015	Burford	N/A	N/A
2016/0151817	12/2015	Nayfeh et al.	N/A	N/A
2016/0151818	12/2015	Shao	N/A	N/A
2016/0167353	12/2015	Fan et al.	N/A	N/A
2016/0175981	12/2015	Kandasamy	N/A	N/A
2016/0175982	12/2015	Kandasamy et al.	N/A	N/A
2016/0184922	12/2015	Kikyo	N/A	N/A
2016/0228932	12/2015	Hayashi et al.	N/A	N/A
2016/0354860	12/2015	Boettcher et al.	N/A	N/A
2017/0008121	12/2016	Li	N/A	N/A
2017/0056947	12/2016	Lavender et al.	N/A	N/A
2017/0136686	12/2016	Ueno et al.	N/A	N/A
2017/0163135	12/2016	Emberton et al.	N/A	N/A
2017/0182587	12/2016	Tokoro et al.	N/A	N/A
2017/0216961	12/2016	Utter et al.	N/A	N/A
2017/0225265	12/2016	Ito et al.	N/A	N/A
2017/0304933	12/2016	Miles et al.	N/A	N/A
2017/0355003	12/2016	Tenhouten et al.	N/A	N/A
2018/0036840	12/2017	Hu et al.	N/A	N/A
2018/0043467	12/2017	Huysmans	N/A	N/A
2018/0050419	12/2017	Das et al.	N/A	N/A
2018/0073532	12/2017	Whalen et al.	N/A	N/A
2018/0311713	12/2017	Joshi et al.	N/A	N/A
2018/0354231	12/2017	Iwase	N/A	N/A
2018/0369889	12/2017	Zhang et al.	N/A	N/A
2019/0267153	12/2018	Kappagantula et al.	N/A	N/A
2019/0275608	12/2018	Das et al.	N/A	N/A
2021/0053100	12/2020	Whalen et al.	N/A	N/A
2021/0086291	12/2020	Okada et al.	N/A	N/A
2021/0205918	12/2020	Fujil et al.	N/A	N/A
2021/0252632	12/2020	Eller et al.	N/A	N/A
2022/0297174	12/2021	Whalen et al.	N/A	N/A
2023/0042802	12/2022	Whalen et al.	N/A	N/A
2023/0078467	12/2022	Komarasamy et al.	N/A	N/A
2023/0081786	12/2022	Joshi et al.	N/A	N/A
2023/0088412	12/2022	Joshi et al.	N/A	N/A
2023/0150022	12/2022	Whalen et al.	N/A	N/A
2023/0234115	12/2022	Kappagantula et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
106140847	12/2015	CN	N/A
107282671	12/2016	CN	N/A
105925846	12/2017	CN	N/A
109295332	12/2019	CN	N/A
112512710	12/2022	CN	N/A
2777837	12/2013	EP	N/A
2990178	12/2017	EP	N/A
3817872	12/2024	EP	N/A
1258141	12/1970	GB	N/A
2002361320	12/2001	JP	N/A
2003275876	12/2002	JP	N/A
2004174563	12/2003	JP	N/A
2007222925	12/2006	JP	N/A
2009090359	12/2008	JP	N/A
2019115909	12/2018	JP	N/A
101316989	12/2012	KR	N/A
WO-2019040730	12/2018	WO	N/A
2020010331	12/2019	WO	N/A
WO-2020053168	12/2019	WO	N/A
WO-2021050022	12/2020	WO	N/A
2021062415	12/2020	WO	N/A
2022043532	12/2021	WO	N/A
2022056358	12/2021	WO	N/A
WO-2023015228	12/2022	WO	N/A
WO-2023043839	12/2022	WO	N/A
2023177693	12/2022	WO	N/A

OTHER PUBLICATIONS

“U.S. Appl. No. 14/222,468, Non Final Office Action mailed Nov. 6, 2015”, 11 pgs. cited by applicant

“U.S. Appl. No. 14/222,468, Office Action mailed Jan. 26, 2017”, 9 pgs. cited by applicant

“U.S. Appl. No. 14/222,468, Office Action mailed May 20, 2016”, 3 pgs. cited by applicant

“U.S. Appl. No. 14/222,468, Office Action, mailed Apr. 1, 2016”, 10 pgs. cited by applicant

“U.S. Appl. No. 14/268,220, Office Action mailed May 2, 2014”, 7 pgs. cited by applicant

“U.S. Appl. No. 17/944,932, Non Final Office Action mailed Apr. 27, 2023”, 19 pgs. cited by applicant

“U.S. Appl. No. 17/944,932, Notice of Allowance mailed Oct. 25, 2023”, 7 pgs. cited by applicant

“U.S. Appl. No. 17/944,932, Response filed Jul. 27, 2023 to Non Final Office Action mailed Apr. 27, 2023”, 10 pgs. cited by applicant

“International Application Serial No. PCT/US2022/043532, International Preliminary Report on Patentability mailed Mar. 28, 2024”, 11 pgs. cited by applicant

“International Application Serial No. PCT/US2022/043532, International Search Report mailed Jan. 12, 2023”, 3 pgs. cited by applicant

“International Application Serial No. PCT/US2022/043532, Written Opinion mailed Jan. 12, 2023”, 6 pgs. cited by applicant

“International Application Serial No. PCT/US2023/015228, International Search Report mailed Jun. 15, 2023”, 5 pgs. cited by applicant

“International Application Serial No. PCT/US2023/015228, Written Opinion mailed Jun. 15,

2023", 6 pgs. cited by applicant

"ThomasNet.com", Online: www.thomasnet.com/articles/custom-manufacturing-fabricating/friction-stir-welding, (Feb. 10, 2011), 2 pgs. cited by applicant

Abu-Farha, Fadi, "A Preliminary Study on the Feasibility of Friction Stir Back Extrusion", *Scripta Materialia* vol. 66 Issue 9; 615-618, (May 2012), 4 pgs. cited by applicant

Amancio-Filho, Sergio T., et al., "Joining of Polymers and Polymer-Metal Hybrid Structures: Recent Developments and Trends", *Polymer Engineering and Science* vol. 49 Issue: 8, (Aug. 2009), 16 pgs. cited by applicant

Bozzi, S., et al., "Intermetallic Compounds in Al 6016/IF-Steel Friction Stir Spot Welds", *Material Science and Engineering: A* vol. 527, Issue: 16-17, (Jun. 25, 2010), 5 pgs. cited by applicant

Cole, G. S., et al., "Lightweight materials for Automotive Applications", *Materials Characterization* vol. 35, Issue: 1, (Jul. 1995), 7 pgs. cited by applicant

Evans, William T., et al., "Friction Stir Extrusion: A new process for joining dissimilar materials", *Manufacturing Letters*, vol. 5; 25-28, (Aug. 2015), 4 pgs. cited by applicant

Gann, John A., "Magnesium Industry's Lightest Structural Metal", *SAE Transactions*, vol. 25/26, (1930-1931), 17 pgs. cited by applicant

Hammond, Vincent H., et al., "Equal-Channel Angular Extrusion of a Low-Density High-Entropy Alloy Produced by High-Energy Cryogenic Mechanical Alloying", *The Journal of the Minerals, Metals & Materials Society*, (Sep. 23, 2014), 9 pgs. cited by applicant

Kaiser, F., et al., "Anisotropic Properties of Magnesium Sheet AZ31", *Materials Science Forum*, vol. 419-4, (2003), 7 pgs. cited by applicant

Kuo, M. C., et al., "Fabrication of High Performance Magnesium/Carbon-Fiber/PEEK/Laminated Composites", *Materials Transactions*, vol. 44, Issue: 8, (2003), 7 pgs. cited by applicant

Leitao, C., et al., "Aluminum-steel lap joining by multi pass friction stir welding", *Materials & Design*, vol. 106, (Sep. 15, 2016), 8 pgs. cited by applicant

Liu, Bin, et al., "Microstructure and mechanical properties of equimolar FeCoCrNi high entropy alloy prepared via powder extrusion", *Intermetallics*, vol. 75,, (Aug. 2016), 6 pgs. cited by applicant

Liu, Liming, et al., "A Review of Dissimilar Welding Techniques for Magnesium Alloys to Aluminum Alloys", *Materials*, (2014), 23 pgs. cited by applicant

Luo, Alan A., "Magnesium: Current and Potential Automotive Applications", *The Journal of the Minerals, Metals & Materials Society*, (Feb. 2002), 7 pgs. cited by applicant

Martinsen, K., et al., "Joining of Dissimilar Materials", *CIRP Annals*, vol. 64, Issue 2, (2015), 21 pgs. cited by applicant

Nakamura, Takashi, et al., "Tool Temperature and Process Modeling of Friction Stir Welding", *Modern Mechanical Engineering*, vol. 8, Issue 1, (Feb. 2018), 17 pgs. cited by applicant

Pickens, J. R., "Aluminum Powder Metallurgy Technology for High-Strength Applications", *Journal of Materials Science*, (Jun. 1981), 21 pgs. cited by applicant

Rodewald, W., et al., "Top Nd—Fe—B Magnets with Greater Than 56 MGOe Energy Density and 9.8 kOe Coercivity", *IEEE Transactions on Magnets*, vol. 38, Issue 5, (Sep. 2002), 3 pgs. cited by applicant

Saha, Pradip K., "Aluminum Extrusion Technology, Chapter 1, Fundamentals of Extrusion", *ASM International*, (2000), 29 pgs. cited by applicant

Trang, T. T.T., et al., "Designing a Magnesium Alloy with High Strength and High Formability", *Nature Communications*, (Jun. 28, 2018), 6 pgs. cited by applicant

Whalen, Scott, et al., "High Ductility Aluminum Alloy Made from Powder by Friction Extrusion", *Materialia*, vol. 6, (Jun. 2019), 6 pgs. cited by applicant

Zhang, Z., et al., "Numerical Studies on Effect of Axial Pressure in Friction Stir Welding", *Science and Technology of Welding and Joining*, vol. 12, Issue 3, (2007), 24 pgs. cited by applicant

"JP 2004-174563 Translation from FIT database", Retrieved from Patent Center with English

translation added, (Year: 2024), 23 pages. cited by applicant
“JP-2002361320-A Translation provided by FIT database”, Retrieved from Patent Center, (Year: 2025), 8 pages. cited by applicant
X, Li, “CN-105925846-B translation provided by FIT database (Year: 2024)”, (Feb. 23, 2018), 12 pgs. cited by applicant

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 17/944,932 filed Sep. 14, 2022, which claims priority to and the benefit of U.S. Provisional Patent Application Ser. No. 63/244,632 filed Sep. 15, 2021, entitled “Co-Extrusion/Cladding of Dissimilar AL Alloys Via Shear Assisted Processing and Extrusion”, the entirety of each of which is incorporated by reference herein.

TECHNICAL FIELD

(1) The present disclosure relates to extrusion assemblies and methods. Particular embodiments include shear-assisted extrusion assemblies and methods that can be used to create multi-metallic materials.

BACKGROUND

(2) Co-extrusion via hydrostatic extrusion and indirect extrusion is either expensive or results in non-uniform material flow. Furthermore, elaborate billet fabrication steps, requirement of specific area ratio between core and the sleeve, difficulty in co-extruding 1100 and 7075AI, and billet preheating are typically employed to obtain desirable bonding at the interface. Thermal spray coatings are limited to thin coatings which might not be beneficial for long-term applications due to cracking and spallation, and explosive bonding has safety and material constraints.

(3) A few research level solutions that can never be applied at a manufacturing scale are fabrication of bi-metallic laminates using equal channel angular pressing and/or high pressure torsion.

(4) Therefore, for increased energy efficiency and reduced plant downtime due to repair, an alternate manufacturing approach is required to fabricate the multi-metallic tubes with relative ease and better interface properties.

SUMMARY

(5) Shear-assisted extrusion assemblies are provided. The assemblies can include: a billet assembly containing a billet comprising a billet outer material and a billet inner material in at least one cross-section; a tool operably engaged with the billet; an extrudate receiving channel configured to receive extrudate from the tool, wherein the extrudate comprises extruded outer material and extruded inner material in at least one cross-section, the extruded outer material being the same material as the billet outer material, and the extruded inner material being the same as the billet inner material.

(6) Methods for producing a multi-material shear-assisted extrudate are also provided. The methods can include: providing a billet comprising billet inner material and billet outer material; providing shear-assisted force to a tool operably engaged with the billet to form an extrudate comprising extruded inner material and extruded outer material wherein the extruded outer material is the same material as the billet outer material, and the extruded inner material being the same as the billet inner material.

Description

DRAWINGS

- (1) Embodiments of the disclosure are described below with reference to the following accompanying drawings.
- (2) FIG. 1 is an example depiction of a portion of a shear-assisted extrusion assembly according to an embodiment of the disclosure.
- (3) FIG. 2 is a cross-section of an extrudate or extruded material according to an embodiment of the disclosure.
- (4) FIG. 3 is an example configuration of a portion of a shear-assisted assembly according to an embodiment of the disclosure.
- (5) FIG. 4 depicts another configuration of a portion of a shear-assisted extrusion assembly according to an embodiment of the disclosure.
- (6) FIG. 5 is a depiction of a shear-assisted extrudate according to an embodiment of the disclosure.
- (7) FIG. 6 is an exploded view of a billet configuration for use with shear-assisted extrusion assemblies according to an embodiment of the disclosure.
- (8) FIG. 7 is an extrudate material according to an embodiment of the disclosure.
- (9) FIGS. 8-12 are example billet configurations and commensurate extrudate materials obtained utilizing the example billet configurations according to embodiments of the disclosure.
- (10) FIG. 13 is a depiction of data of extruded material according to an embodiment of the disclosure.
- (11) FIG. 14 is a depiction of the operating conditions of a shear-assisted extrusion assembly utilizing differing materials according to an embodiment of the disclosure.
- (12) FIG. 15 is another example of shear-assisted extrusion assembly parameters according to an embodiment of the disclosure.
- (13) FIG. 16 is an example of co-metallic extruded material properties according to an embodiment of the disclosure.
- (14) FIGS. 17A-17E are depictions of multi-metallic extruded material according to an embodiment of the disclosure.
- (15) FIGS. 18A-18C are depictions of interfacial portions of multi-metallic extruded materials according to an embodiment of the disclosure.
- (16) FIGS. 19A-19D are depictions of multi-metallic extruded materials according to an embodiment of the disclosure.
- (17) FIGS. 20A-20E are depictions of interfacial multi-metallic extruded materials according to an embodiment of the disclosure.
- (18) FIGS. 21A-21C are depictions of portions of multi-metallic extruded materials according to an embodiment of the disclosure.
- (19) FIGS. 22A-22D are depictions of multi-metallic extruded materials according to an embodiment of the disclosure.
- (20) FIGS. 23A-23B are depictions of a billet configuration according to an embodiment of the disclosure.
- (21) FIGS. 24A-24B are depictions of billet configurations according to an embodiment of the disclosure.
- (22) FIGS. 25A-25C are depictions of multi-metallic extruded materials according to an embodiment of the disclosure.
- (23) FIGS. 26A-26B are depictions of multi-metallic extruded materials according to an embodiment of the disclosure.
- (24) FIG. 27 is a depiction of a multi-metallic extrudate according to an embodiment of the disclosure.

(25) FIG. **28** is a depiction of a multi-metallic extrudate according to an embodiment of the disclosure.

DESCRIPTION

(26) This disclosure is submitted in furtherance of the constitutional purposes of the U.S. Patent Laws “to promote the progress of science and useful arts” (Article 1, Section 8).

(27) The present disclosure will be described with reference to FIGS. **1-28**. The present disclosure provides assemblies and methods for creating a co-extruded multi-metallic extrudates that can be co-extruded in the form of tubes. The extrudates can include at least an extruded inner material within an extruded outer material. The inner material can be referred to as a core and the outer material can be referred to as a cladding or sleeve. In tube configurations, the inner material can be referred to as an inner layer and the outer material as an outer layer.

(28) The materials (inner, outer, intermediate) that make up the extrudate can have distinct chemical and/or mechanical properties. For example, co-extrusion of 6061 (shell) and 7075 (core), 1100 (shell) and 7075 (core), and 1100 (shell) and 2024 (core) was performed to complete extrudates having commensurate outer and inner materials having sound bonding at the interface. Furthermore, thickness of the sleeve and core can be controlled via the area ratio of the constituent billet material. Multi-metallic tubes can be fabricated utilizing a tool that engages a mandrel.

(29) The ShAPE process works in billet area ratio of $1/3$ and $2/3$, and also $1/2$ and $1/2$ leading to the tube thickness of $1/3$ and $2/3$, and $1/2$ and $1/2$, respectively. Based on the current observations, a much thinner core/sleeve, preferably having $1/4$ of the tube thickness is quite possible. Therefore, either a thicker or thinner coating can easily be fabricated. Temperature control of the process using the shear and friction parameters can provide for more flexible fabrication of multi-metallic components having significantly different flow stresses. Furthermore, multi-metallic extrudate tubes exhibiting variable thicknesses can be fabricated by controlling the billet stack with minimal effort in billet fabrication. This allows for joining the bi-metallic tubes to other structures so that easily weldable material of necessary thickness is the sleeve. Difficult to extrude material combination such as 1100 and 7075/2024 AI can easily be co-extruded.

(30) Furthermore, these multi-metallic extrudates, having gone through high temperature severe plastic deformation, can be aged without solutionizing or annealing heat treatments, thereby thermal stresses and the subsequent dimensional instability can be avoided and/or grain growth minimized.

(31) Referring to FIG. **1**, a portion of an example shear-assisted extrusion apparatus **10** is shown that includes a billet holder assembly **12** that comprises and container base **12a** and a container ring **12b**. Within assembly **12** is billet **20**. Billet **20** comprises at least two materials, but may contain additional materials in different alignments. In this configuration, billet material **20** comprises at least a billet inner material **22b** and a billet outer material **24b**. As an example, materials **22b** and **24b** will be different materials as noted herein. The materials may differ in chemical and/or mechanical properties. The materials may also have different dimensions. For example, the thickness ratio of the materials as shown appear to be the same. Other thickness ratios can be utilized with the outer material being less or greater thickness than the inner material.

(32) Tool **14** can be retained within tool holder **21** and operably engaged with billet material **20** to create a high shear region **26** at die face **28**. As shown, a rotational force and axial force is applied to die face **28** of tool **14**. The axial force may be applied from the tool upon the feed material; alternatively, the axial force may be applied from the feed material upon the tool. In accordance with example implementations, the shear assisted extrusion process will provide an extrudate **18**. Mandrel **16** can be anchored in billet holder assembly **12** and extend through billet material **20** and die face **28** to extrudate receiving channel **11**.

(33) This application is related to U.S. Continuation-In-Part patent application Ser. No. 17/473,178 filed Sep. 13, 2021, entitled “Devices and Methods for Performing Shear-Assisted Extrusion and Extrusion Processes”. the entirety of which is hereby incorporated by reference herein.

- (34) Referring to FIG. 2, extrudate **18** can comprise at least extruded inner material **22e** and extruded outer material **24e**. These extruded materials (**22e/24e**) can be the same as billet materials (**22b/24b**) and have substantially the same ratio of thickness (e.g., area ratio of 1/3 and 2/3, and also 1/2 and 1/2 leading to the tube thickness of 1/3 and 2/3, and 1/2 and 1/2, respectively). The extruded materials can be bonded at interface **23** while the billet materials are not bonded.
- (35) Referring to FIGS. 3-12, various combinations of billet material configurations as well as extruded material configurations are shown. For example, FIGS. 3-7 demonstrate inner billet material **22b** and outer billet material **24b** bounded by billet material **40b**. When shear-assisted extruded, extruded inner material **22e** and outer extruded outer material **24e** are bounded by extruded material **40e**. This extruded material can have bonding interfaces between the inner **22e** and outer **24e** materials as well as between the outer **24e** and extruded material **40e** and/or the inner **22e** and extruded material **40e**.
- (36) Referring to FIGS. 8-12 a summary of example billet and extruded material configurations are shown. This summary is not provided to be exhaustive, just as an example. Accordingly, the billet material and the extrudate can include intermediate materials **90b/90e**. These intermediate materials can include multiple materials themselves. Thus, the materials can include multiple layers in at least one cross-section. Additionally, with reference to FIGS. 10 and 11, inner and/or outer materials can extend to complete and be the same material as bounding materials. Accordingly, the inner or outer materials can be the same as bounding materials (e.g., **40**). Also, with reference to FIG. 12, multiple inner and multiple outer materials can be provide as shown. Accordingly, outer materials can include distinct materials **24** and **120**. Additionally, inner materials can include distinct materials **22** and **122**. As the depiction provides, the abutment of these materials need not align.
- (37) Accordingly, billet containing assembly **12** containing a billet that includes a billet outer material **22b** and a billet inner material **24b** in at least one cross-section. The billet **20** can also include a billet intermediate material **90b** between billet inner material **22b** and billet outer material **24b** in the at least one cross-section. The assembly can include mandrel **16** extending from billet containing assembly **12** to tool **14**. Mandrel **16** can extend through an opening in die face **28** of the tool **14**. Billet material **20** can also include a billet boundary material **40b** directly adjacent and/or lateral of either billet inner material **22b** or outer material **24b**. the billet boundary material is the same material as the billet outer material. Billet boundary material **40b** can be a different material than either or both of the billet inner material **22b** or outer material **24b**. In at least one cross-section the thickness of one or more of the inner **22e**, outer **24e**, and/or intermediate **90b** billet materials is different. Each material thickness can be as low as a 10.sup.th of the thickness of the material directly adjacent thereto. In billet container assembly **12**, the billet materials can be unbound.
- (38) The extrudate **18** can also include extruded intermediate material **90e** between extruded outer material **24e** and extruded inner material **22e** in the at least one cross-section.
- (39) In accordance with the above assemblies, the methods can include a billet intermediate material **90b** between the billet inner material **22b** and billet outer material **24b**. Billet intermediate material **90b** can be a single material or multiple different materials. Inner **22b** and outer **24b** billet material can be about the same thickness in at least one cross-section. Extruded inner **22e** and outer **24e** materials can be about the same thickness in at least one cross-section. Inner billet material **22b** can be 1/10 to 9/10 the thickness of outer **24b** billet material. Extruded inner material **22e** can be 1/10 to 9/10 of the thickness of extruded outer billet material **24e**. The extruded materials define bonded interfaces between different and previously unbound billet materials.
- (40) As an example method with reference to Table 1 below, multi-metallic extrudates were prepared. These parameters are not limiting and ranges of rotational speed as well as advance rate can be utilized.
- (41) TABLE-US-00001 TABLE 1 Summary of example shear-assisted processing conditions

employed to fabricate the multi-material extrudates. Material Die advance Tube combination speed (RPM) rate (mm/min) length, mm 1100 (24) and 40 60 1960 7050 (22) Al 1100 (24) and 45 1830 2024 (22) Al

(42) The extruded tubes were sectioned for various microscopy and mechanical property characterizations in as-extruded condition. For optical and scanning electron microscopy (SEM) analysis, the tubes were sectioned along both longitudinal and transverse cross-sections. The cut samples were mounted in an epoxy, polished to a surface finish of 0.05 μm using colloidal silica suspension, and etched using Keller's reagent. Both stereo and light microscopy (in bright field mode) analyses of the samples were carried to analyze the tube quality and interface characteristics. The samples were repolished to 0.05 μm surface finish for SEM and energy-dispersive X-ray spectroscopy (EDS) analysis. SEM in back scattered electron (BSE) mode and EDS analyses of the clad tubes and the interface between 1100/7075 Al, and 1100/2024 Al were completed using the FEI Quanta 3D field emission gun dual beam FIB/SEM. Vickers hardness measurements in as-extruded was carried out using Clark Microhardness tester CM-700AT and five indents per sample was done. Tensile testing of 127 mm (5") long tubes was carried out in accordance with ASTM-B557 standard. All the results and discussion are presented in the next section.

(43) Referring to FIG. 3, and with reference to Table 2, tensile properties of multi-metallic extrudates having differing inner and outer thickness ratios are shown.

(44) TABLE-US-00002 TABLE 2 Material combinations YS (MPa) UTS (MPa) EI % 6061/7075 Al 175 252 13 ($\frac{1}{2}$ and $\frac{1}{2}$) 6061/7075 Al 168 224 8.5 ($\frac{2}{3}$ and $\frac{1}{3}$) 1100/7075 Al 132 219 9 1100/2024 Al 116 198 11.5

(45) The measured tool temperature ($^{\circ}\text{C}$.) and the spindle torque (Nm) as a function of plunge depth for both the extrusions is presented in FIGS. 13 and 14, respectively. What can be considered a long steady-state temperature region was followed by an initial transient temperature region.

(46) Steady-state temperatures in 1100/7075 Al and 1100/2024 Al were about 380 and 390 $^{\circ}\text{C}$., respectively, mainly due to slightly lower rotational speed in the former extrusion. An initial transition in spindle torque was also noted (FIG. 15) followed by an almost steady-state. However, beyond plunge depth of ~ 40 mm, oscillation in torque was observed. Exact origin of these oscillations is not known; however, a possible hypothesis could be based on the sticking characteristics of 1100 Al exhibits during thermomechanical processing. During ShAPE process, 1100 Al sticks to the extrusion die and at higher force the material slips, then the continuation of this process would result in torque oscillations, as noted in FIG. 15.

(47) Referring next to FIG. 16, a comparison of coextruded multi-metallics is shown. Additional analysis of coextruded materials have been performed using Stereo and light microscopy of clad 1100 on 7075 Al and 1100 on 2024 Al are presented in FIGS. 17A-17E and 20A-20E respectively. FIGS. 17A-17B and 20A-20B represent the macro-view of both longitudinal and transverse cross-sections for clad 1100/7075 Al and 1100/2024 Al, respectively. The clad outer layer (1100 Al) was continuously present along the tube (dark region) in all the conditions and cross-sections analyzed. The bright inner region in FIGS. 17A-17B and 20A-20B correspond to 7075 Al and 2024 Al, respectively. Thickness of tubes measured on the transverse cross-section for 1100/7075 and 1100/2024 are 0.92 ± 0.05 mm and 0.97 ± 0.14 mm, respectively. As noted, tube thickness in 1100/2024 Al was higher than 1 mm in certain locations as compared with 1100/7075 Al. The thickness of the outer and inner layers for 1100/7075 Al are 0.6 ± 0.1 and 0.36 ± 0.06 mm, respectively, and thickness for the outer and inner layers for 1100/2024 Al are 0.57 ± 0.1 and 0.38 ± 0.03 mm, respectively. In both the tubes, the 1100 Al outer layer exhibited larger thickness than the expected 0.5 mm based on the billet area. The light microscopy images are presented in FIGS. 17B-17E and 20B-20E. Both outer and inner layers of the tube for both clad configurations did not have any processing flaws. Furthermore, higher magnification analysis of the interface region shows that the interface in both the conditions is defect-free and wavy in nature.

(48) SEM-BSE and SEM-EDS analyses of 1100/7075 and 1100/2024 Al clad tubes are presented in FIGS. **18A-18C** and **21A-21C**, respectively. Interface region between 1100/7075 and 1100/2024 Al clad tubes is marked with a solid arrow. For the most part, interface region did not exhibit discontinuities except for a few micro-cracks as marked by the dotted arrow in FIG. **18A**. Therefore, the interface region was metallurgically bonded in both the conditions as clearly observed in FIGS. **18B** and **21B**. Furthermore, presence of second phases is noted in both 7075 and 2024 Al alloys. Results of SEM-EDS line scan analysis of 1100/7075 Al and 1100/2024 Al are presented in FIGS. **18C** and **21C**, respectively. As expected, Zn and Cu content increased across the clad interface from 1100 Al into the 7075 and 2024 Al, respectively. Additional data are provided in Tables 3-4 below regarding hardness and tensile strength of the multi-metallic materials.

(49) TABLE-US-00003 TABLE 3 Hardness summary in as-extruded condition for 1100 and 7075 Al and 1100 and 2024 Al clad tubes Material Hardness identification (HV0.1) 1100/7075 Al 1100 Al 20.2 ± 0.2 7075 Al 83.3 ± 3.0 1100/2024 Al 2024 Al 68.0 ± 1.0 1100 Al 21.0 ± 0.3

(50) TABLE-US-00004 TABLE 4 Tensile properties Summary of the clad Al tubes. YS (MPa) UTS (MPa) EI. % 1100/7075 Al 135 ± 3.5 213 ± 5 6.4 ± 1.4 1100/2024 Al 117 ± 5.4 199 ± 4 10.6 ± 1.1

(51) Overall, the presence of metallurgically bonded interface obtains clad tubes with good structural integrity.

(52) Referring next to FIGS. **19A-19D** and **22A-22D**, inner and outer materials having similar cross-section area are provided and metallurgical bonding was determined to be present in both cases of 1100 outer and 7075 or 2024 inner.

(53) In accordance with an example implementation, at least a pair of billet designs are shown in FIGS. **23A-23B** and **24A-24B**. FIGS. **23A-23B** demonstrate a similar thickness while FIGS. **24A-24B** demonstrate an inner 1/3 to outer 2/3 ratio. The materials used were inner 7075 and outer 6061. In FIGS. **25A-28** analysis results of both FIGS. **23A-B** and **24A-B** designs are shown.

(54) Successful extrudate fabrication of distinctly different flow strength materials can provide heat exchangers with high temperature strength materials having a corrosion/oxidation resistance material as either an inner or outer material. At least one example of this application is co-extrusion of copper-nickel bi-metallic tubing which has been performed using these methods. Moreover, the apparatus and methods described herein enable continuous cladding from start to stop. For example, the outer material can be present over an entirety of an extrudate surface. Additionally, or alternatively, the apparatus and method provide the ability to vary or change outer material thickness. That is, a thicker and/or thinner outer material may be continuously fabricated along an outer surface of the extrudate.

(55) In compliance with the statute, embodiments of the invention have been described in language more or less specific as to structural and methodical features. It is to be understood, however, that the entire invention is not limited to the specific features and/or embodiments shown and/or described, since the disclosed embodiments comprise forms of putting the invention into effect. The invention is, therefore, claimed in any of its forms or modifications within the proper scope of the appended claims appropriately interpreted in accordance with the doctrine of equivalents.

Claims

1. A shear assisted extrusion assembly, the assembly comprising: a billet containing assembly configured for receiving a billet comprising a billet outer material and a billet inner material in at least one cross-section; a tool configured to operably engage the billet outer material and the billet inner material to cause plastic deformation of the billet outer material and the billet inner material; and an extrudate receiving channel configured to receive extrudate from the tool, the extrudate comprising extruded outer material and extruded inner material in at least one cross-section;

wherein the extruded outer material comprises the same material as the billet outer material, and the extruded inner material comprises the same material as the billet inner material.

2. The assembly of claim 1 wherein the extruded inner material and the extruded outer material are bonded at an interface therebetween.
3. The assembly of claim 2 wherein: further comprising the billet; and wherein the billet further comprises a billet intermediate material between the billet outer material and the billet inner material in the at least one cross-section; and the extrudate further comprises extruded intermediate material between the extruded outer material and the extruded inner material in the at least one cross-section.
4. The assembly of claim 1 further comprising a mandrel extending from the billet containing assembly to the tool, wherein the mandrel extends through an opening in a die face of the tool.
5. The assembly of claim 1 further comprising the billet; and wherein the billet further comprises a billet boundary material adjacent to either the billet inner material or the billet outer material.
6. The assembly of claim 5 wherein; the billet boundary material is axial to the billet inner material; and the billet boundary material is axial to the billet outer material.
7. The assembly of claim 5 wherein the billet boundary material is a different material than the billet outer material.
8. The assembly of claim 5 wherein the billet boundary material comprises a different material than either or both of the billet inner or outer materials.
9. The assembly of claim 5 wherein in at least one cross-section a thickness of the inner and outer billet materials is different.
10. The assembly of claim 1, further comprising the billet.
11. A method for producing a multi-material shear assisted extrudate, the method comprising: positioning a billet to be operably engaged with a tool to cause plastic deformation of the billet, the billet comprising a billet inner material and a billet outer material; and establishing rotation and axial force between the tool and the billet to engage the tool with the billet to form an extrudate comprising extruded inner material and extruded outer material; wherein the extruded outer material and the extruded inner material are bonded at an interface therebetween.
12. The method of claim 11 wherein the extruded outer material comprises the same material as the billet outer material and the extruded inner material comprises the same as the billet inner material.
13. The method of claim 11 further comprising a billet intermediate material between the billet inner material and the billet outer material, wherein the billet intermediate material comprises a single material.
14. The method of claim 13 wherein the billet intermediate material comprises multiple different materials.
15. The method of claim 11 wherein the inner and outer billet material are about a same thickness in at least one cross-section.
16. The method of claim 15 wherein the extruded inner material and the extruded outer material are about the same thickness in at least one cross-section.
17. The method of claim 11 wherein the inner billet material is 1/10 to 9/10 the thickness of the outer billet material.
18. The method of claim 17 wherein the extruded inner material is 1/10 to 9/10 of the thickness of the extruded outer billet material.
19. The method of claim 11 wherein the extruded inner and outer materials define bonded interfaces between different unbound billet materials.
20. A shear assisted extrusion assembly, the assembly comprising: a billet containing assembly containing a billet, the billet comprising a billet outer material and a billet inner material in at least one cross-section; a tool configured to operably engage the billet outer material and the billet inner material to cause plastic deformation of the billet outer material and the billet inner material; a mandrel extending from the billet containing assembly to the tool, wherein the mandrel extends

through an opening in a die face of the tool; and an extrudate receiving channel configured to receive extrudate from the tool, the extrudate comprising extruded outer material and extruded inner material in at least one cross-section; wherein the extruded inner material and the extruded outer material are bonded at an interface therebetween.
